II. Subordinate projections and partial order on P(A)/~

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II Subordinate projections and partial order on $\mathcal{P}(A)/\sim$

The relation $\sim_A$ on $\mathcal{P}(A)$ may be seen as

$$P \sim_A Q \overset{(*)}{\iff} P \preceq Q \text{ and } Q \preceq P$$

for the partial order $\preceq$ on $\mathcal{P}(A)$, defined as

$$P \preceq Q \overset{\text{def.}}{\iff} \exists R \in A \text{ s.t. } P \sim_A R \text{ and } R \leq Q.$$

If $P \preceq Q$, we say P is <u>subordinate</u> to Q.

(Intuitively, $P(H)$ can be mapped isometrically into $Q(H)$.
Note that $P \leq Q$ implies $P \preceq Q$ (just take $R = P$).)

Proof of equivalence $(*)$:

$(\Rightarrow)$ For $P \preceq Q$, take $R := Q$. For $Q \preceq P$, $R := P$.

$(\Leftarrow)$ $\exists Q_1, P_1 \in A$ s.t.

- $P \sim_A Q_1$, $Q \sim_A P_1$
  - ↳ I.e. $\exists U, V \in A$: $P = U^*U$, $Q_1 = UU^*$
    $Q = V^*V$, $P_1 = VV^*$
- $Q_1 \leq Q$, $P_1 \leq P$.

Trick: define

$$P_2 = V Q_1 V^* \quad \left(P_1(H) \xrightarrow{V^*} Q(H) \xrightarrow{Q_1} Q_1(H) \xrightarrow{V} \text{subset of } P_1(H) \right)$$

$$Q_2 = U P_1 U^* \quad \left(Q_1(H) \to \text{subset of } Q_1(H) \right)$$

$\vdots$

$$\left.\begin{aligned} P_n &= V Q_{n-1} V^* \\ Q_n &= U P_{n-1} U^* \end{aligned}\right\} (*)$$

This way, we get two sequences of inequalities

$$P_0 := P \geq P_1 \geq P_2 \geq \cdots$$

$$Q_0 := Q \geq Q_1 \geq Q_2 \geq \cdots$$

(Proven easily by induction)

Defining $P_\infty := \bigwedge_{n \in \mathbb{N}} P_n$ and $Q_\infty := \bigwedge_{n \in \mathbb{N}} Q_n$,

(i.e. projections to $\bigcap_n P_n(H)$ and $\bigcap_n Q_n(H)$)

and noting that

$$P_\infty = \lim_{n \to \infty}^{SOT} P_n, \quad Q_\infty = \lim_{n \to \infty}^{SOT} Q_n$$

we get (by applying the strong limit to $(*)$)

$$U P_\infty U^* = Q_\infty, \quad V Q_\infty V^* = P_\infty$$

and clearly,

$$U (P_n - P_{n+1}) U^* \overset{\text{def. of } P_n, Q_n}{=} Q_{n+1} - Q_{n+2}$$

$$V (Q_n - Q_{n+1}) V^* = P_{n+1} - P_{n+2}.$$

This means

$$P_n - P_{n+1} \sim_A Q_{n+1} - Q_{n+2}$$

$$Q_n - Q_{n+1} \sim_A P_{n+1} - P_{n+2}$$

$$P_\infty \sim_A Q_\infty$$

and so

$$P = \sum_{n=0}^{\infty}{}^{SOT}\left(P_n - P_{n+1}\right) + P_\infty$$

$$= \underbrace{\sum_{n=0}^{\infty}{}^{SOT}\left(\underbrace{P_{2n} - P_{2n+1}}_{\sim_A\, Q_{2n+1} - Q_{2n}}\right)} + \underbrace{\sum_{n=0}^{\infty}{}^{SOT}\left(\underbrace{P_{2n+1} - P_{2n+2}}_{\sim_A\, Q_{2n} - Q_{2n+1}}\right)} + \underbrace{\overbrace{P_\infty}^{\sim Q_\infty}}$$

3 pairwise orthogonal proj's.

$\sim_A Q$

Note that this is possible since

$$\left(P_n - P_{n+1}\right) \perp \left(P_{n+2} - P_{n+3}\right)$$

and we may apply the proposition from before. $\blacksquare$

It's not hard to show that if $P, Q, T \in \mathcal{P}(A)$, then $P \preceq Q$ and $Q \preceq T$ implies $P \preceq T$.

Because of this (and the previous proposition), $\preceq$ passes to the quotient $\mathcal{P}(A)/\sim_A$, i.e.

$$[P] \preceq [Q] \overset{\text{def}}{\iff} P \preceq Q$$

is well-defined. Furthermore, this relation is transitive and

$$\left([P] \preceq [Q] \text{ and } [Q] \preceq [P]\right) \iff [P] = [Q]$$

so that $\preceq$ is a partial order on $\mathcal{P}(A)/\sim_A$.

Separable

<u>Example</u>. If P_1 and P_2 are proj's in $B(H)$, then
$$P_1 \preceq P_2 \iff \dim(P_1(H)) \leq \dim(P_2(H)).$$

($\Rightarrow$) Trivial.

($\Leftarrow$) Pick ON bases $(e_i)_{i\in I_1}$, $(f_i)_{i\in I_2}$ for $P_1(H)$ and $P_2(H)$, $I_1 \subset I_2$. Map:
$$V : H \to H, \quad \sum_{i\in I_1}' x_i e_i \mapsto \sum_{i\in I_2}' x_i f_i$$
and $V|_{P_1(H)^\perp} := 0$.